

SESSION 3

Ground To Sky

A Deep Dive into MAVLink

Industrial Exposure Session · Open-Source Autopilot Systems
EDYODA

AGENDA

Session Overview

- **Introduction to MAVLink**

What it is, why it exists, and its role in UAV communication.
Interactive: “Who talks to whom?” activity.

- **MAVLink Architecture & Message Format**

Understand how MAVLink packets are structured and transmitted.
Visual: Packet breakdown, message fields, CRC.

- **Communication Flow in UAV Systems**

How GCS, Flight Controller, and Companion Computer exchange data.
Diagram: Multi-link MAVLink network.

- **Hands-On: Sending and Receiving MAVLink Messages**

Using pymavlink to monitor telemetry and send commands.
Live demo: HEARTBEAT, ATTITUDE, COMMAND_LONG.

- **Advanced Topics — MAVLink v2 & Security Enhancements**

Message signing, CRC extra, custom XML messages.
Showcase: Custom message generation demo.

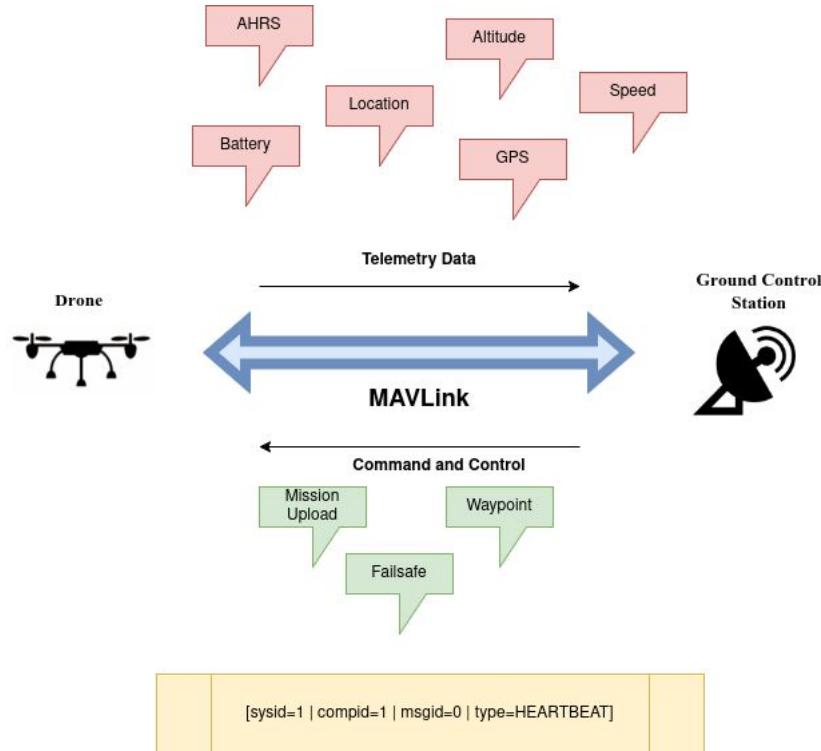
Goal

Understand MAVLink architecture, message flow, and practical implementation for UAVs



Speaking the Drone's Language

Understanding How Drones, Ground Stations, and Computers Communicate in the Sky



Lightweight Design

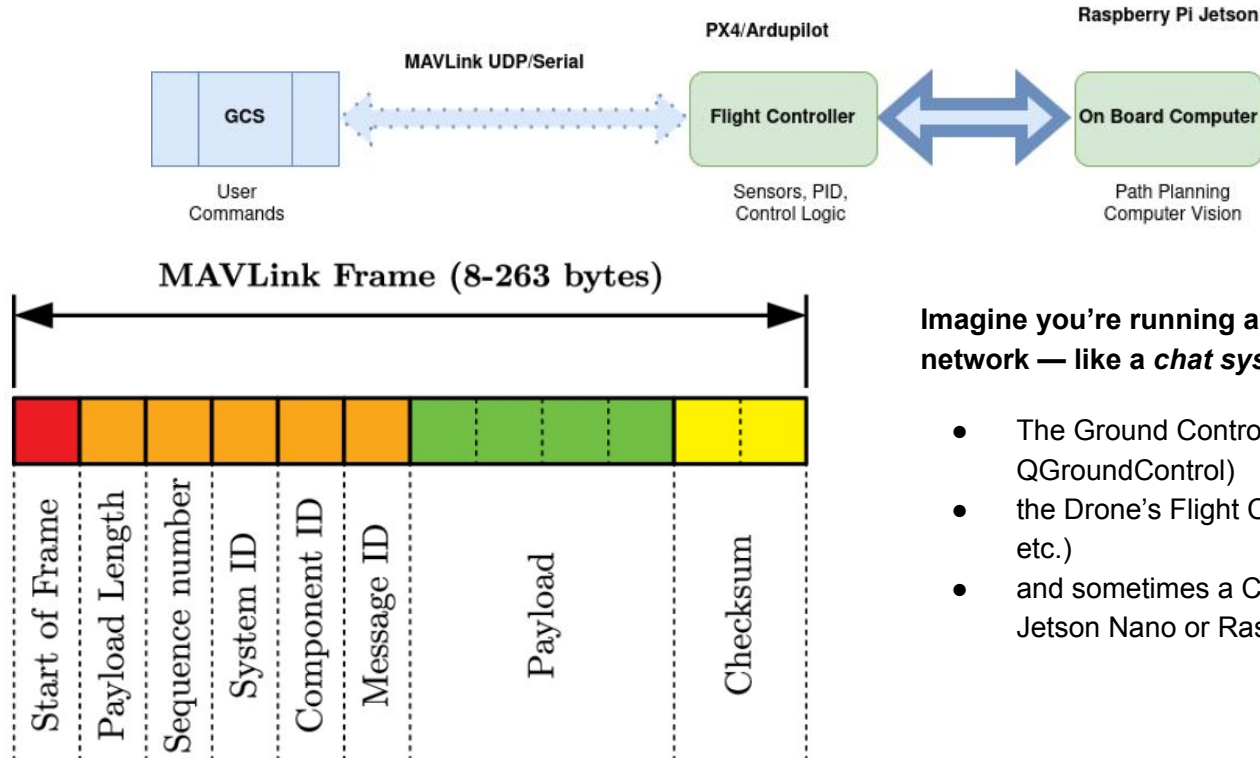
Optimized for micro-controllers with minimal memory footprint and processing overhead



Bidirectional Communication

Supports both telemetry (vehicle to GCS) and command (GCS to vehicle) data exchange

MAVLink Message Format — The Anatomy of a Packet



Imagine you're running a drone communication network — like a *chat* system between:

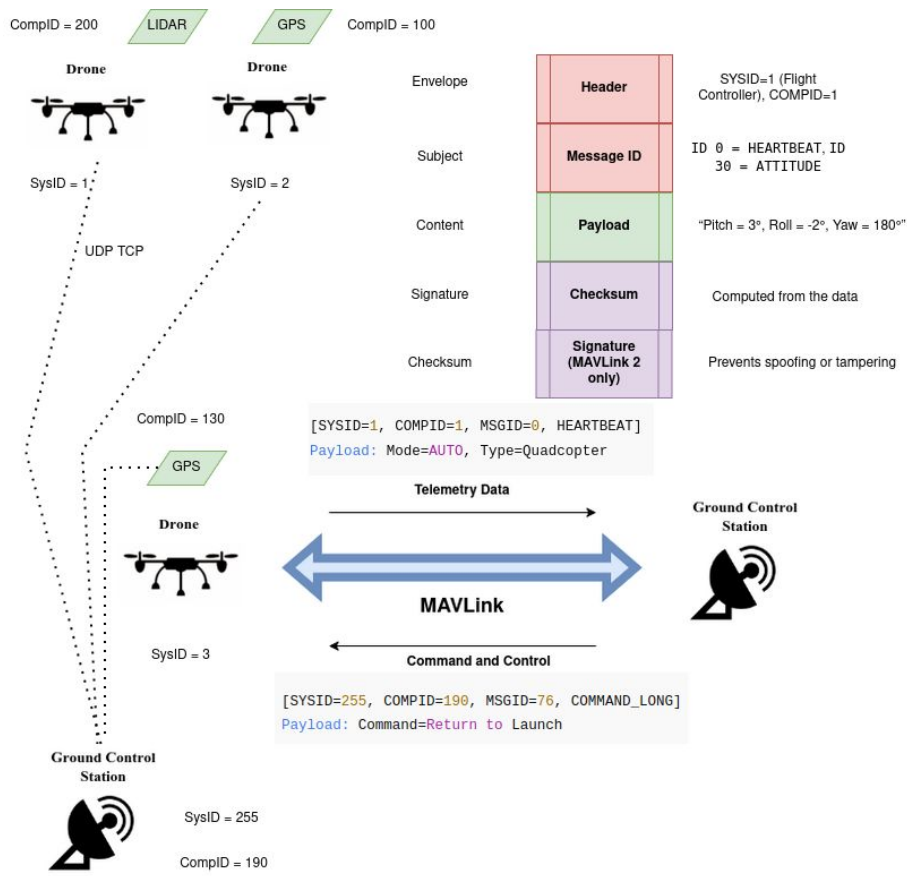
- The Ground Control Station (GCS) (like QGroundControl)
- the Drone's Flight Controller (like Pixhawk, Cube, etc.)
- and sometimes a Companion Computer (like Jetson Nano or Raspberry Pi).

Understanding MAVLink Packet Structure — Simplified Explanation

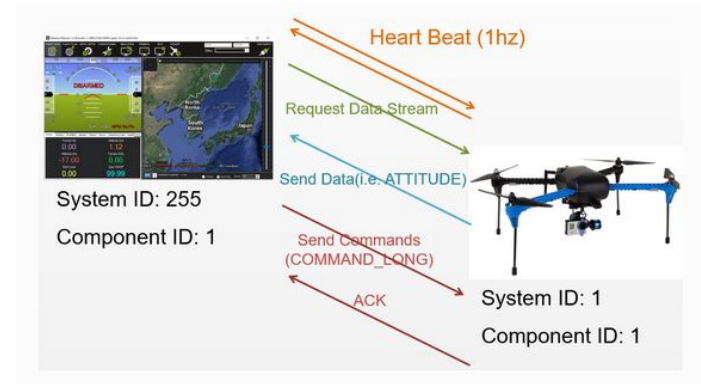
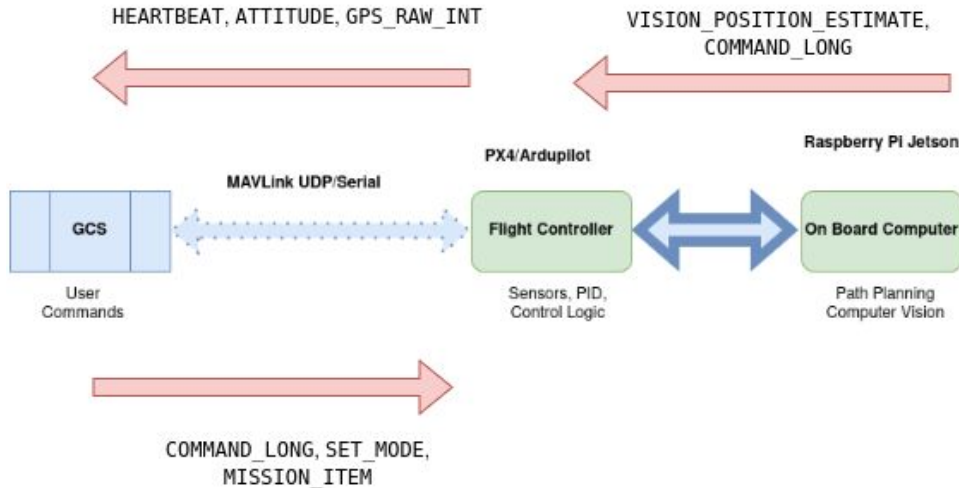
Example analogy:

Imagine you're sending a letter to your friend:

- The envelope (**header**) has your name and address.
- The subject (**message ID**) says what the letter is about.
- The content (**payload**) is your actual message.
- The signature proves *you* wrote it.
- The **checksum** ensures it wasn't changed in transit.



MAVLink Communication Flow



Quick Analogy

"Think of MAVLink as a group chat

The Ground Station says **'Take off!'**

The Flight Controller replies **'Altitude 20m reached!'**

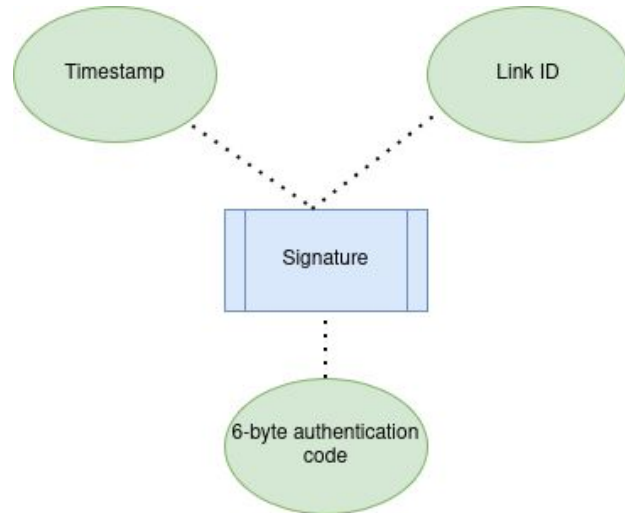
The Companion Computer adds **'Object detected ahead.'**

All in the same language — **MAVLink.**"

Advanced Concepts — MAVLink 2: Security

Concept

MAVLink 2 can *sign each message* with a digital fingerprint — proving it really came from your drone or GCS.



🧱 Purpose:

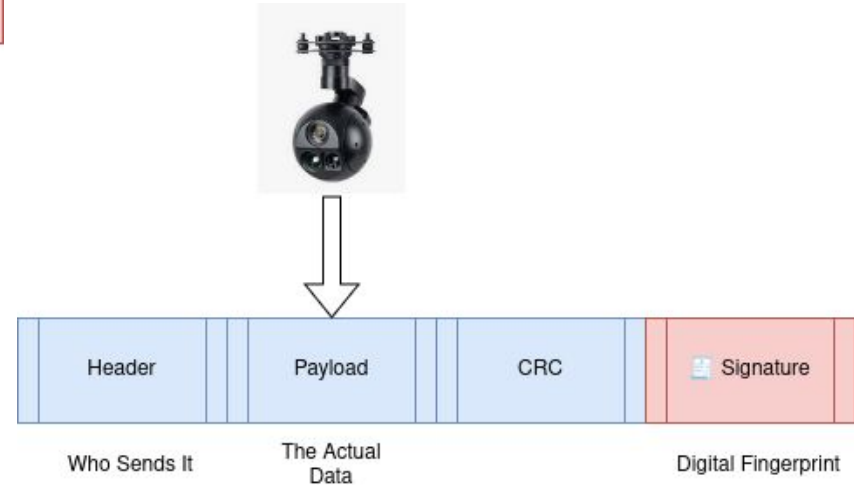
- Prevent **spoofing** (fake messages)
- Stop **replay attacks** (resending old commands)
- Protect **command integrity** in multi-drone or defense setups

Advanced Concepts — MAVLink 2 Extensions



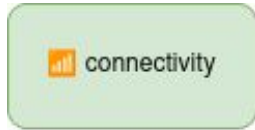
Each message is signed like a digital handshake — verifying the sender and preventing spoofing.

Feature	MAVLink 1	MAVLink 2
Message Size	263 bytes	280 bytes
Extensions	✗	✓
Message Signing	✗	✓
Backward Compatible	✗	✓
CRC Extra	Basic	Stronger Check

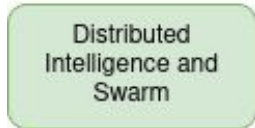


Extension : Future-proof your drones without breaking old systems.

The Future of UAV Communication



MAVLink 2 isn't just communication — it's the digital backbone of next-gen UAV networks.

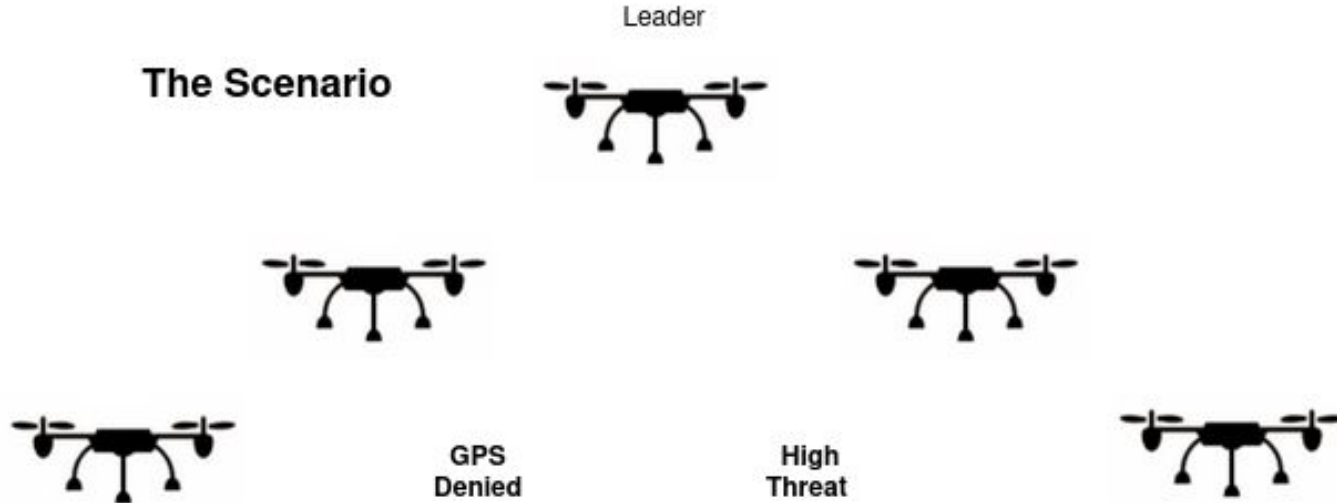


Trend	What It Means
Encrypted MAVLink	Research for full message encryption (beyond signing)
DroneCAN Integration	MAVLink over CAN for modular avionics
Swarm & AI Extensions	Data sharing between autonomous UAVs
Standardized Extensions	Common fields for industry compatibility

Collaborative Combat Aircraft

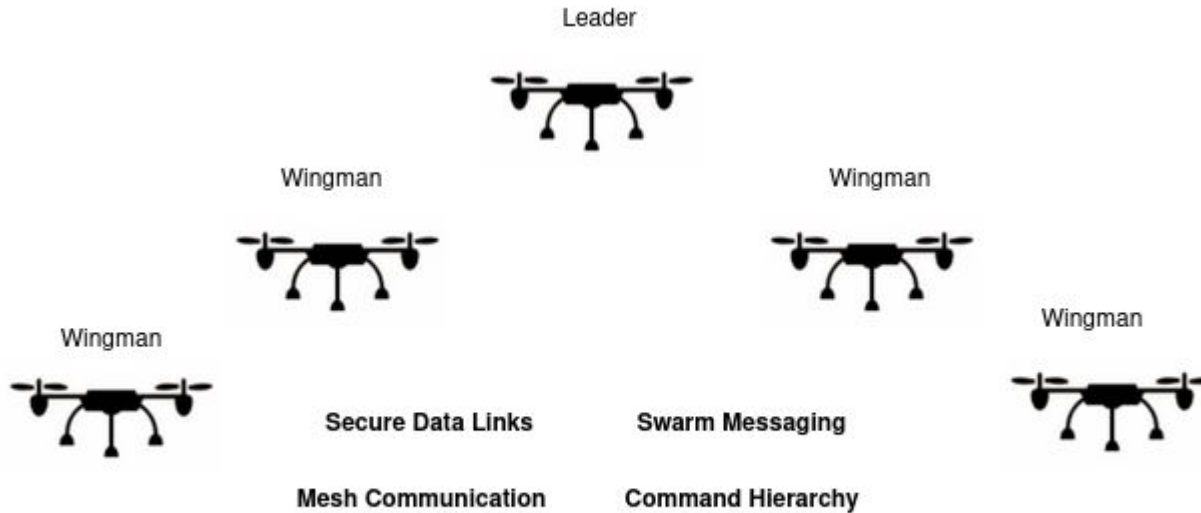
Case Study : How MAVLink 2 Powers Real Time Coordination in Next Generation combat UAV

Mission Scenario



Each UAV is both a sensor and a teammate — sharing data, reacting collectively.

MAVLink 2: The Digital Backbone



MAVLink 2 enables the swarm to talk — safely and smartly.

Key Technical Highlights

 Security

- Message signing ensures command authenticity
- CRC checks protect message integrity

 Network

- Multi-node UDP link with dynamic routing
- Fallback to serial or radio if link lost

 Extensibility

- Custom `CCA_STATUS` message (e.g., target vector, threat level, Threat prioritization)
- Payload module updates over MAVLink

Capability	Enabled by MAVLink 2
Real-time target sharing	Message broadcasting
Secure coordination	Message signing
Modular payload upgrades	Message extensions
Reduced GCS dependency	Peer-to-peer links

Collaborative Combat Aircraft (CCA) – MAVLink Based Case Study Questions

Q1.

Design a MAVLink-based communication architecture for a manned fighter aircraft controlling:

- 4 kamikaze UAVs
- 2 ISR drones
- 1 EW drone

How would you:

- assign system IDs,
- manage telemetry bandwidth,
- prioritize mission commands,
- ensure fail-safe operation?
- What if Leader Fails

Q2.

How would you implement a decentralized swarm architecture using MAVLink where UAVs can continue operation even if:

- the Leader link is lost,
- one drone fails,
- GPS is jammed?

Explain:

- message routing,
- leader election,
- heartbeat handling,
- autonomy layers.

Q3.

A high-speed CCA UAV operating at 40 m/s in acro/stabilize mode experiences telemetry latency spikes during terrain masking.

How would you redesign:

- MAVLink transport
- Command acknowledgement?